

# Homological Methods in Commutative Algebra

## Problem Set 2

Throughout, all rings are commutative and noetherian, and all modules are finitely generated.

**Problem 1.** Let  $I$  be a nonzero proper ideal in a noetherian domain  $R$  and let  $f_1, \dots, f_c$  be a maximal regular sequence inside  $I$ . Consider the short exact sequence

$$0 \longrightarrow N \longrightarrow R/(f_1, \dots, f_c) \xrightarrow{\pi} R/I \longrightarrow 0.$$

where  $\pi$  is the canonical quotient map.

- a) Show that  $\text{Ext}_R^{c-1}(N, R) = 0$ .
- b) Show that the following induced map on  $\text{Ext}$  is nonzero:

$$\pi^* = \text{Ext}_R^c(\pi, R) : \text{Ext}_R^c(R/I, R) \longrightarrow \text{Ext}_R^c(R/(f_1, \dots, f_c), R).$$

**Problem 2.** Let  $(R, \mathfrak{m})$  be a noetherian local ring and take elements  $f_1, \dots, f_c \in I$ . Consider the Koszul complex  $E = \text{kos}(f) = \bigwedge(e_1, \dots, e_c)$  and let  $F$  be a dg  $R$ -algebra resolution of  $R/I$  with  $F_0 = R$ . Fix  $a_1, \dots, a_c \in F_1$  such that  $\partial_F(a_i) = f_i$ . We define a map of complexes  $\varphi : E \rightarrow F$  as follows: we set  $\varphi_0 = \text{id}_R$  and for each  $1 \leq s \leq c$ , consider the  $R$ -module map  $\varphi_s : E_i \rightarrow F_s$  given on basis elements by

$$\varphi_s(e_{i_1} \wedge \dots \wedge e_{i_s}) = a_{i_1} \cdots a_{i_s}.$$

- a) Show that  $\varphi \partial = \partial \varphi$ , that is, that  $\varphi$  is a map of complexes.
- b) Show that for all  $z \in \ker(\varphi)$  and all  $w \in E$ ,  $wz \in \ker(\varphi)$ .

Let  $k$  be a field and let  $f_1, \dots, f_n$  be monomials in  $k[x_1, \dots, x_d]$ , minimally generating the ideal  $I = (f_1, \dots, f_n)$ . The Taylor resolution of  $R/I$  has a dg algebra structure, defined on basis elements as follows:

$$e_J \cdot e_L = \text{sgn}(J, L) \frac{f_J f_L}{f_{J \cup L}} b_{J \cup L}.$$

Here the sign is given by

$$\text{sgn}(J, L) = (-1)^\varepsilon \quad \text{where} \quad \varepsilon = |\{(j, \ell) : j \in J, \ell \in L, \text{ and } j > \ell\}|$$

and by convention, if  $J \cap L \neq \emptyset$ , then  $\text{sgn}(J, L) = 0$ .

**Problem 3.** Let  $I = (f_1, \dots, f_n)$  be a monomial ideal in  $Q = k[x_1, \dots, x_d]$  and consider its Taylor resolution  $T$ . Fix a subset  $J \subseteq [n]$ .

- a) Show that

$$\partial(e_J) \notin \mathfrak{m}T \iff f_i \mid f_{J \setminus \{i\}} \text{ for some } i \in J.$$

- b) Show that for each  $i \in [n]$ ,

$$e_i e_J \notin \mathfrak{m}T \iff \gcd(f_i, f_j) = 1 \text{ for all } j \in J.$$

**Problem 4.** Let  $k$  be any field and  $R = k[x, y, z_1, \dots, z_{n-1}]$  for some  $n \geq 3$ . Show that the ideal

$$I = \left( x^n, y^n, \sum_{i=1}^{n-1} z_i x^i y^{n-i} \right)$$

has  $\text{pdim}(R/I) = n + 2$ .